

Parallel Tempered MCMC (PTMCMC)

Drawbacks of regular MCMC:

- We don't know convergence time (auto correlation length)
- We can't measure the evidence

Parallel tempering speeds up convergence and gives us access to evidence.

We have many chains, each with "temperature"
 T_i . $\beta_i = 1/T_i \in [0, 1]$

Instead of likelihood, we use

$$p(\mathcal{D}|\theta) \rightarrow p(\mathcal{D}|\theta)^{\beta_i}$$

Cold chain ($\beta=1$)

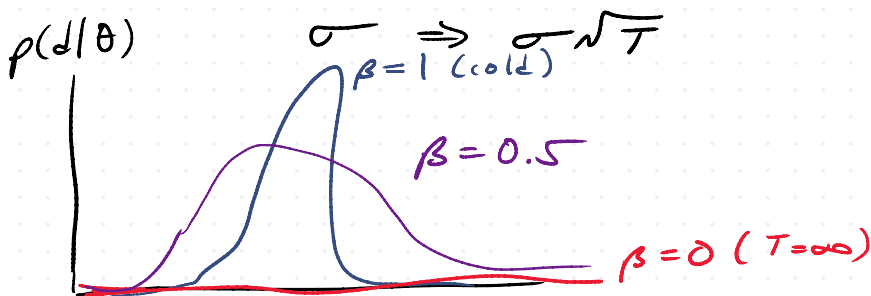
$$p(\theta|\mathcal{D}) \propto p(\mathcal{D}|\theta)^{\beta} \cdot p(\theta)$$

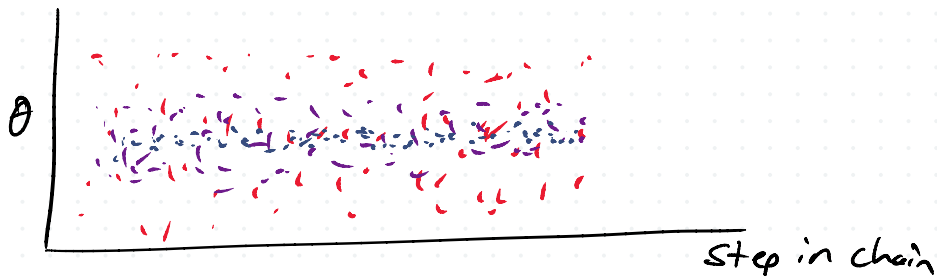
Hot chain ($\beta=0, T=\infty$)

$$p(\theta|\mathcal{D}) \propto p(\theta)$$

$$p(\mathcal{D}|\theta) \sim e^{-\chi^2}$$

$$p(\mathcal{D}|\theta)^{\beta} \Rightarrow e^{-\chi^2 \beta} = e^{-\chi^2 / T}$$





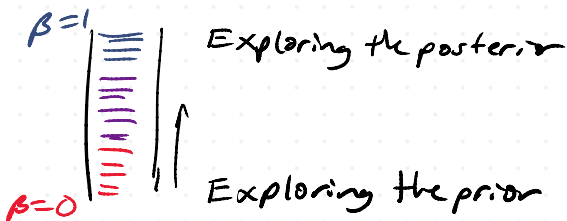
PTMCMCs are very good for when likelihood has multiple peaks (multimodal distributions)

$$\alpha(\beta_i \longleftrightarrow \beta_j) = \min \left[1, \left(\frac{p(d|\theta_j)}{p(d|\theta_i)} \right)^{\beta_i} \cdot \left(\frac{p(d|\theta_i)}{p(d|\theta_j)} \right)^{\beta_j} \right]$$

The algorithm:

- Pick a number of temperatures and spacing of temperatures (temperature ladder)
 - - Run each one independently as an MCMC for x steps. Can be in parallel!
 - Run temperature swap check.
- ↗ Swap w/ prob. $\alpha(\beta_i, \beta_j)$

Which T_i, T_j you compare and how often you do T-swaps are up to you!



One temperature ladder "geometric spacing"

$$T_i = c^i \quad (\text{sometimes } c=2)$$

Another option: evenly spaced in β

Evidence in PTMCMC

WARNING! You probably want to use a nested sampler in practice.

WARNING! Thermodynamic integration sucks!
Better method: stepping stone method.

Thermodynamic Integration

Say we have two unnormalized distributions

q_0, q_1

$$p_i(\theta) = \frac{1}{Z_i} q_i(\theta)$$

$$Z_i = \int q_i(\theta) d\theta$$

We want to evaluate the log ratio

$$\mu = \log(Z_1/Z_0) = \log Z_1 - \log Z_0$$

Define a continuous differentiable path

$$q_0 \rightarrow q_1, \quad q_\beta, \quad 0 \leq \beta \leq 1$$

$$p_\beta(\theta) = \frac{1}{Z_\beta} q_\beta(\theta)$$

$$Z_\beta = \int q_\beta(\theta) d\theta$$

$\log Z_\beta$, take a derivative:

$$\begin{aligned}
 \frac{\partial}{\partial \beta} \log Z_\beta &= \frac{1}{Z_\beta} \frac{\partial Z_\beta}{\partial \beta} \\
 &= \frac{1}{Z_\beta} \frac{\partial}{\partial \beta} \int q_\beta d\theta \\
 &= \frac{1}{Z_\beta} \int \frac{\partial q_\beta}{\partial \beta} d\theta \\
 &= \frac{1}{\int q_\beta d\theta} \cdot \int \frac{\partial q_\beta}{\partial \beta} d\theta \\
 &\stackrel{(\text{?})}{=} \int \underbrace{\frac{1}{q_\beta} \frac{\partial q_\beta}{\partial \beta}}_{\frac{\partial \log q_\beta}{\partial \beta}} \underbrace{\left[\frac{q_\beta}{Z} \right]}_{p_\beta} d\theta
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial \log Z_\beta}{\partial \beta} &= \int \frac{\partial \log q_\beta}{\partial \beta} \cdot p_\beta d\theta \\
 \int_0^1 \frac{\partial \log Z_\beta}{\partial \beta} d\beta &= \int \left\langle \frac{\partial \log q_\beta}{\partial \beta} \right\rangle d\beta
 \end{aligned}$$

$$\langle U \rangle_{\beta} \equiv \left\langle \frac{\partial \log q_{\beta}}{\partial \beta} \right\rangle$$

$$\log Z_1 - \log Z_0 = \int_0^1 \langle U \rangle_{\beta} d\beta$$

For PTMCMC, $q_0 = p(\theta)$ prior

$$q_1 = p(d|\theta) p(\theta) \quad \text{un-normalized posterior}$$

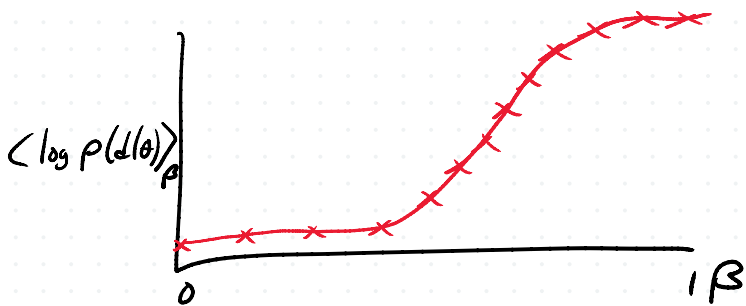
$$q_{\beta} = p(d|\theta)^{\beta} p(\theta)$$

$$Z_0 = 1, \log Z_0 = 0.$$

$$Z_1 = \text{evidence.}$$

$$\frac{\partial}{\partial \beta} \log q_{\beta} = \beta \cdot \log p(d|\theta) + \frac{\partial \log p(\theta)}{\partial \beta}$$

$$\log p(d) = \int_0^1 \langle \log p(d|\theta) \rangle_{\beta} d\beta$$



Recommended reading:

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